AMTI (NMTC) - 2012

RAMANUJAN CONTEST - INTER LEVEL

PART - A

1. The sum of the squares of all real numbers satisfying the equation $x^{256} - 256^{32} = 0$ is
(A) 8 (B) 128 (C) 512 (D) 65536

2. Two adjacent vertices of a square are on a circle of radius R and the other two vertices lie on a tangent to the circle. The length of the side of the square is
(A) $\frac{4R}{3}$ (B) $\frac{6R}{5}$ (C) $\frac{8R}{5}$ (D) $\frac{3R}{2}$

3. The polynomial $x^{2n} + 1\ (x + 1)^{2n}$ is not divisible by $(x^2 + x + 1)$ if n is equal to
(A) 17 (B) 20 (C) 21 (D) 64

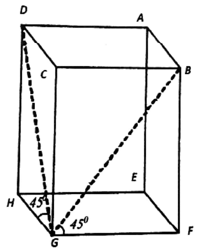
4. The remainder when 3^{302} is divided by 11 is
(A) 10 (B) 9 (C) 8 (D) 7

5. If a, b and d are the lengths of a side, a shortest diagonal and a longest diagonal of a regular nonagon, then
(A) $d^2 = a^2 + ab + b^2$ (B) $d^2 = a^2 + b^2$
(C) $d = a + b$ (D) $b^2 = ad$

6. If a positive integer, after adding 100, becomes a perfect square, and also after adding 168, becomes another perfect square. (168 is added to the original positive integer), then the sum of the digits of the integer is
(A) 12 (B) 11 (C) 8 (D) 7

7. ABC is a triangle inscribed in a circle. AD is the altitude of the triangle. DP is drawn parallel to AB to cut the tangent at A at P. Then $\angle CPA$ is
(A) Equal to 90^0 when triangle ABC is acute angled
(B) Equal to 90^0 for any non-right angled triangle ABC
(C) Greater than 90^0 for the angle A obtuse
(D) Smaller than 90^0 for an acute angled triangle ABC

8. In the adjoining figure of a rectangular solid, it is given that $\angle DGH = 45^0$ and $\angle BGF = 60^0$. Then $\cos(\angle BGD) =$



(A) $\frac{\sqrt{6}-\sqrt{2}}{4}$ (B) $\frac{\sqrt{2}}{6}$ (C) $\frac{\sqrt{6}}{4}$ (D) $\frac{\sqrt{3}}{6}$

9. For the inequation $(1.25)^{1-x} < (0.64)^{2(1+\sqrt{x})}$.
(A) There is no real value of x exists
(B) There are infinitely many negative value of x exist
(C) Any real value of x greater than 25 satisfies the inequation
(D) There is exactly one positive and one negative integer only satisfying the equation

10. The number $(5^6 - 10^4)$ is
(A) Negative (B) divisible by 10 (C) divisible by 100 (D) divisible by 9

11. There are 'a' points on a line and 'b' points on a parallel line. The number of triangles formed by these (a + b) points as vertices is
(A) $\frac{ab(a+b-2)}{2}$ (B) $\frac{ab(a+b-1)}{2}$ (C) $\frac{ab(a+b-4)}{2}$ (D) $\frac{ab(a+b)}{2}$

12. The equation $x^4 + 16x - 12 = 0$ has
(A) all the roots real
(B) all the roots complex
(C) two roots real and two roots complex
(D) all the roots integers

13. A positive integer n has 60 divisors and 7n has 80 divisors. What is the greatest value of k such that 7^k divides n.
(A) 1 (B) 2 (C) 3 (D) 0

14. In the XY plane two points A(2, 2) and B(7, 7) are taken. R is the region in the first quadrant which consists of points C such that triangle ABC is an acute angled triangle. The closest integer to the area of the region R is
(A) 25 (B) 39 (C) 51 (D) 60

15. How many three digit numbers have distinct digits such that one digit is the average of the other two ?
(A) 104 (B) 112 (C) 256 (D) 12

PART - B

1. Let $f(x) = x^2 + bx + c$ where b, c are integers. If f(x) is a factor of both $x^4 + 6x^2 + 25$ and $3x^4 + 4x^2 + 28x + 5$, then the value of f(1) is _______.

2. Mahadevan's age is 'a' years, which is also the sum of the ages of his three children. His age b years ago was twice the sum of their ages. Then $\left(\frac{a}{b}\right)$ = _______.

3. $\angle$RST is an angle in the minor segment of a circle of centre O, then the angle (in degrees) RST less the angle ORT is _______.

4. A, B, C are three towns connected by straight roads from A to B, B to C and C to A. AB = 5 km, BC = 6 km, CA = 7 km. Two cyclists start simultaneoulsy form A and go in different roads with same speed. They meet at D. then BD = _______.

5. f is a linear function given by f(x) = ax + b and $f^{-1}(x)$ = bx + a when a, b are real. The value of a + b is equal to ________.

6. ABC is triangle. A point O is taken inside the triangle such that $\angle BOC = 120^0$. OD, OE, OF are drawn perpendiculars to the sides BC, CA, AB respectively. Then $\angle$EDF + EAF = ______.

7. ABC is an isosceles triangle in which AB = BC. BC is produced to D such that $\angle CAD = \frac{1}{2}\angle BAC$. If L is the foot of the $\perp$r from C to AD, then the value of $\frac{AL}{AD}$ = ____.

8. The values of x satisfying the inequality $x - \sqrt{1-|x|} < 0$ lie in $\left[-1, \frac{a}{2}\right]$. Then te value of a is ______.

9. For some real numbers a and b, the equation $8x^3 + 4ax^2 + 2bx + a = 0$ has three distinct positive roots. If the sum of the logarithms to base 2 of the roots is 5, the value of a is _____.

10. ABCD is a quadrilateral in the first equation of the coordinate axes. A is (3, 9), B is (1, 1), C is (5, 3) and D is (a, b). The quadrilateral formed by joining the mid-points of AB, BC, CD and DA is a square. The sum of the coordinates of the point D is ______.

11. The number of distinct four-tuples of numbers (a, b, c, d) of rational numbers satisfying $a\log_{10}2 + b\log_{10}3 + c\log_{10}5 + d\log_{10}7 = 2012$ is ______.

12. If the graph of $f(x) = ||x - 2| - a| - 3$ has exactly three x – intercepts, then a is equal to ____.

13. If we add the square of the digit in the tens place of a positive two digit number to the product of the digits of the number, we get 52. If we add the square of the digit in the units place of the number to the same product of the digits, we get 117. The two digit number is _______.

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